# NIKHIL REDDY LEVAKU

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## PROFESSIONAL SUMMARY

Backend Engineer building secure RESTful APIs with **Core Java**, **Spring Boot**, and **PostgreSQL** across the full software development lifecycle (SDLC). Experienced in microservices architecture, stateless **JWT/RBAC** security, **Redis** caching, and event-driven asynchronous messaging with **Apache Kafka** and **RabbitMQ**. Strong grounding in transactional concurrency, query optimization, automated testing with JUnit 5 and Mockito, and Agile sprint delivery.

## TECHNICAL SKILLS

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| **Languages & Frameworks** | Java, SQL, **Spring Boot**, **Spring Security**, Hibernate/JPA |
| **Databases** | PostgreSQL, MySQL, Redis, Flyway |
| **Messaging & Architecture** | Apache **Kafka**, **RabbitMQ**, Microservices, REST API Design |
| **Testing & Tools** | JUnit 5, Mockito, Postman, Swagger/OpenAPI, Git, GitHub, Gradle, Maven |
| **Concepts** | JWT, RBAC, Transaction Management, Concurrency, Event-Driven Design, Agile/Scrum |
| **Cloud & DevOps** | Docker, Docker Compose, Kubernetes (Deployments, Services), Terraform (Modules), Nginx, Jenkins (CI), GitHub Actions (CI), SonarQube, SonarScanner, Bash, Linux, AWS (EC2, VPC, S3) |

## PROFESSIONAL EXPERIENCE

**DeepLure Research Pvt Ltd — Backend Engineer** **May 2026 – Aug 2026**

Java · Spring Boot · Spring Security · PostgreSQL · Hibernate/JPA · Redis · RabbitMQ

- Analyzed business requirements and shipped **100+ production REST APIs** across B2B and on-demand platforms, building configurable backend components for authentication, orders, payments, and admin modules.
- Secured APIs with **Spring Security**, stateless **JWT**, and **RBAC**, with OTP verification over SMS and email.
- Integrated the **Razorpay** payment gateway (orders, wallets, refunds), using idempotency keys and concurrency locking to prevent double-charges and race conditions under parallel requests.
- Built asynchronous notification workflows with **RabbitMQ**; managed application secrets with **Infisical**.
- Built real-time support ticketing and chat over **WebSocket/STOMP**, with **Firebase Cloud Messaging** push notifications and third-party SMS integration.
- Built **Redis-backed asynchronous workers** for notifications, payment reconciliation, and booking expiry; reduced database load **~40%** through Redis caching and query optimization.
- Modeled normalized **PostgreSQL** schemas with Hibernate/JPA and versioned **Flyway** migrations; aligned API contracts with frontend teams via Swagger/OpenAPI; wrote **JUnit 5**/**Mockito** tests and fixed defects in Agile sprints.

**SmartBridge Educational Services (APSCHE) — Data Analytics Intern** **May 2025 – Jul 2025**

- Cleaned and transformed toy-manufacturing sales datasets using SQL joins, filters, and aggregate queries, and built 3 Tableau dashboards for state-level production and inventory review.

## PROJECTS

**Flight Booking System — Distributed Microservices Platform** **Aug 2025 – Jan 2026**

Java · Spring Boot · Spring Security · Microservices · PostgreSQL · Redis · Apache Kafka · Docker · Nginx | GitHub

- Refactored a monolithic backend into **4 independently deployable microservices** (Auth, Flight, Booking, Notification), with per-service **PostgreSQL** databases and **Flyway** migrations for data isolation.
- Implemented inter-service REST communication with stateless **JWT** propagation, so each service independently authenticates requests and enforces **RBAC** (USER/ADMIN).
- Prevented overbooking under concurrent requests with transactional seat reservation using row-level **pessimistic write locking** (**PESSIMISTIC_WRITE**) inside **@Transactional** methods; added seat rollback logic.
- Built an **event-driven** pipeline with **Apache Kafka**: Booking Service publishes booking-created events consumed asynchronously by the Notification Service.
- Added **Redis caching** for flight search, request validation, centralized exception handling, and **JUnit 5**/**Mockito** tests.
- Containerized services with multi-stage **Docker** builds and ran the full stack with **Docker Compose** behind an **Nginx** reverse proxy; added a **GitHub Actions** CI workflow and deployed to **AWS EC2**.

**User Management System** **Oct 2024 – Dec 2024**

Java · Spring Boot · Hibernate/JPA · MySQL · REST API | GitHub

- Built a CRUD REST API with dynamic search and filtering, request validation, and centralized exception handling.
- Modeled a normalized 6-table MySQL schema with JPA entity mappings; tested and documented all endpoints in Postman.

## EDUCATION

**Sri Venkateswara College of Engineering**, Tirupati **2022 – May 2026**

B.Tech, Electronics & Communication Engineering CGPA: 8.5 / 10

## CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate — Oracle (Sep 2025) · Problem Solving, Intermediate — HackerRank (Apr 2026) · SQL, Intermediate — HackerRank (Mar 2026) · Java Programming — GeeksforGeeks (Mar 2026)